DAM SAFETY MONITORING

Expert Reference: dam types, failure modes, monitoring parameters, codes, IoT sensors, data, thresholds and alerts

Knowledge foundation for the DAM MASTER AI digital tool

Item	Detail
Document ID	RCPL-DMAI-REF-001
Version	1.0 (Draft for expert review)
Date	10 October 2026
Prepared by	Digital vertical, Rodic Hub — Rodic Consultants Pvt. Ltd.
Readers	Management, dam safety engineers, instrumentation engineers, product and development team
Status	Reference document. Standard editions, clauses and all numeric thresholds must be confirmed by the dam safety engineer for each dam.

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1. The big picture — monitoring step by step

A dam is monitored the way a doctor looks after a patient with a known condition: first understand the patient (the dam), list what could go wrong (failure modes), decide which vital signs reveal those problems early (parameters), fit the right instruments (sensors), know what "normal" looks like (baseline), and act at agreed warning levels (alerts). The 12 steps below are the backbone of this document and of the digital tool.

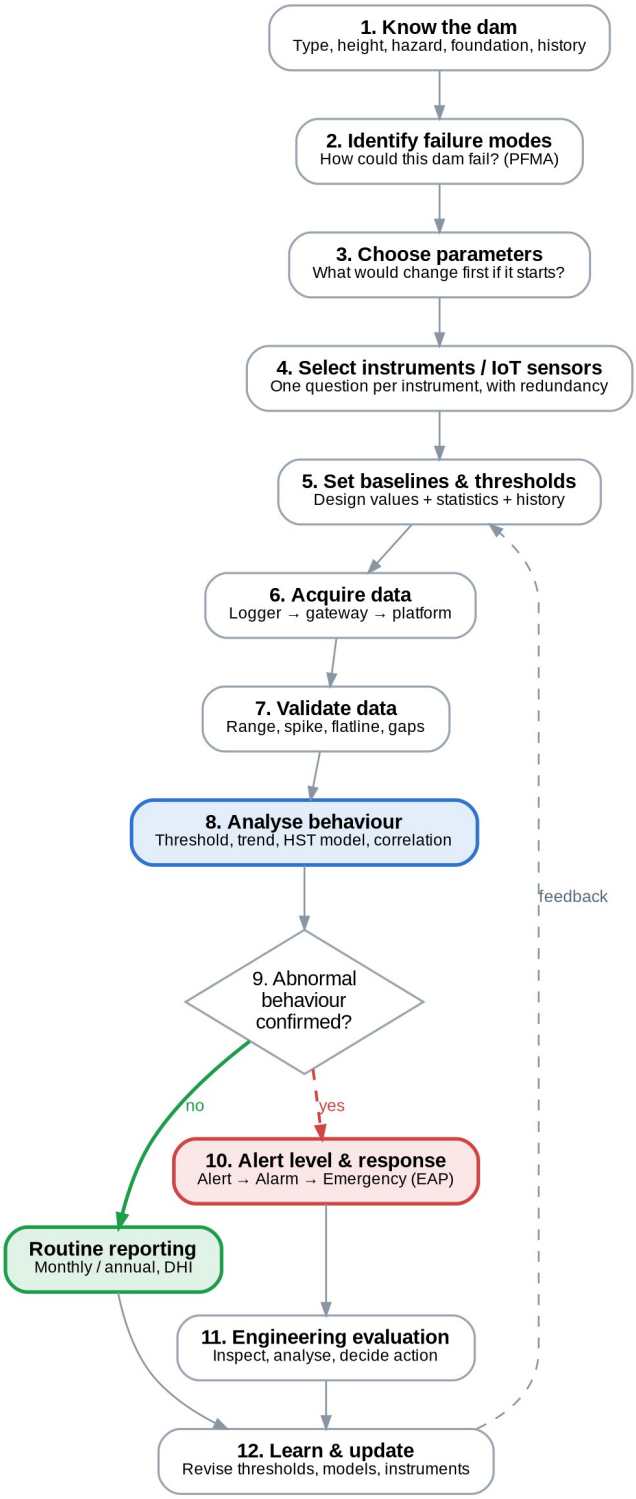


Figure 1.1 — Dam safety monitoring, step by step

Step	Question it answers	Chapter
1. Know the dam	What kind of dam is it, how high, what is downstream?	2

Step	Question it answers	Chapter
2. Failure modes	How could this particular dam fail?	3
3. Parameters	What changes first if a failure starts?	4
4. Codes	Which standards and rules apply?	5
5. Sensors	Which instruments measure those parameters?	6
6. Data	What does each sensor send, how often, in what form?	7
7–8. Analysis	How do we decide if behaviour is normal?	8
9. Meaning of changes	If a reading rises by X, what could be happening?	9
10. Alerts	When must an alert go out, to whom, how fast?	10
11–12. Inspection and improvement	How do inspections, EAP and records fit in?	11–12
Digital tool	How all this becomes software	13

> Golden rule of instrumentation (after Ralph Peck): every instrument must be there to answer a specific question about a specific failure mode. An instrument with no question is cost without safety.

2. Types of dams

Dams are classified in four independent ways. The digital tool stores all four for every dam because each one changes what is monitored and how strict the alerts are.

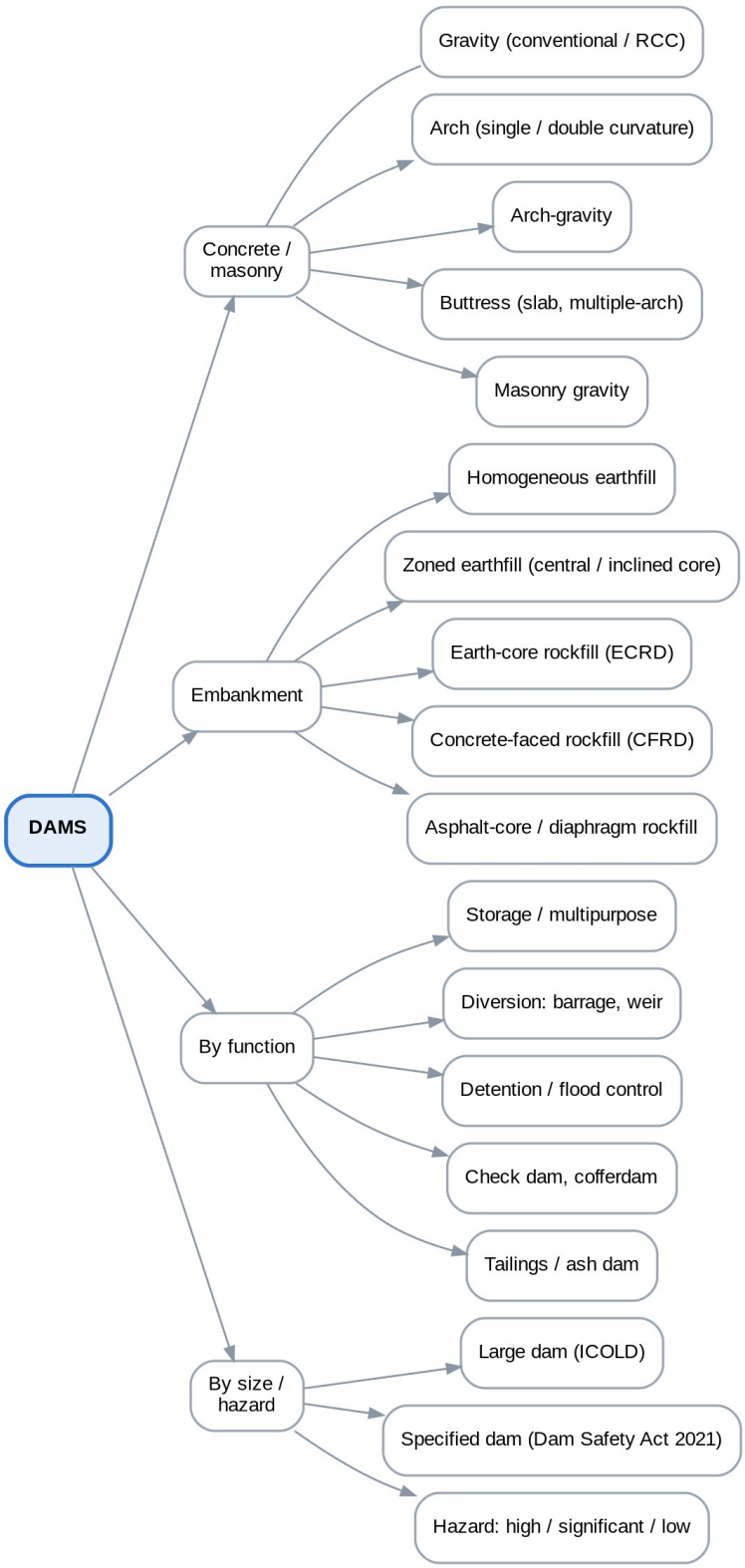


Figure 2.1 — Dam classification

2.1 By material and structure

Type	How it holds water (plain English)	Where used	Main weaknesses to watch	Monitoring priority
Concrete gravity (conventional)	Its own weight resists the water push, like a heavy block	Rock foundations; spillway on the dam body	Sliding on foundation, uplift under the base, cracking, joint opening, ageing concrete	Uplift pressure, drain flow, pendulum / crest movement, joints, seepage, temperature

Type	How it holds water (plain English)	Where used	Main weaknesses to watch	Monitoring priority
Roller-compacted concrete (RCC) gravity	Same as gravity, built faster in thin compacted layers	Large gravity dams where speed and cost matter	Seepage along lift joints, thermal cracking	Seepage per gallery zone, lift-joint wetness, temperature, uplift
Arch (single or double curvature)	Curved shell carries water load sideways into the valley walls	Narrow gorges with strong rock abutments	Abutment rock movement, cracking from thermal and load cycles	Pendulums, abutment extensometers, joint meters, strain, temperature, seismic
Arch-gravity	Part arch, part weight	Moderately narrow valleys	Combination of both	As arch plus uplift
Buttress (flat slab / multiple arch)	Thin upstream face supported by a row of buttresses	Older dams; where concrete saving mattered	Buttress cracking, lateral stability in earthquakes, face slab deterioration	Crack meters, tilt, seismic, concrete condition
Masonry gravity	Stone masonry in mortar acting by weight	Many older Indian dams	Mortar leaching, seepage through masonry, uplift	Seepage quantity and chemistry, uplift, visual leaching
Homogeneous earthfill	One soil type; water seeps slowly through it	Low dams (typically up to about 30 m)	Internal erosion (piping), downstream slope failure, rising phreatic line	Piezometers, seepage and turbidity, settlement, downstream wet areas
Zoned earthfill (central or inclined core)	Clay core stops water; sand/gravel shells give stability; filters protect the core	Medium to high dams on varied foundations	Core cracking, filter failure, piping, settlement, rapid-drawdown slope failure	Core and foundation piezometers, seepage, turbidity, settlement cells, inclinometers
Earth-core rockfill (ECRD)	Clay core with rockfill shells	High dams in rock valleys	Core hydraulic fracturing, differential settlement, core arching	Pore pressure in core, settlement, inclinometers, total pressure cells, seepage
Concrete-faced rockfill (CFRD)	Rockfill body; concrete slab on upstream face is the water barrier	Narrow to wide rock valleys, seismic areas	Face slab cracking, perimeter joint opening, leakage through slab	Perimeter joint meters (3-D), slab strain, rockfill settlement, seepage weir
Asphalt-core / diaphragm rockfill	Thin asphalt or concrete wall inside the fill is the barrier	Where clay is unavailable	Core cracking or joint leaks	Seepage, core deformation, settlement

2.2 By function

Type	Purpose	Monitoring note
Storage / multipurpose	Store water for irrigation, drinking water, power, flood control	Full instrumentation; long-term behaviour models
Barrage / weir	Raise river level for diversion; little storage; gates control flow	Gates, scour downstream, uplift under floor, piping under foundation
Detention / flood control	Hold floods temporarily; often empty	Behaviour during rare fillings is critical — instruments must work after long idle periods
Check dam, cofferdam	Small or temporary structures	Overtopping and erosion; often visual plus water level
Tailings / ash dam	Store mining or power-plant waste	Raised in stages; high liquefaction and piping risk; pore pressure and phreatic level critical

2.3 By size and hazard

Classification	Rule	Why it matters
Large dam (ICOLD)	Height ≥ 15 m from lowest foundation to crest, or 5–15 m impounding more than 3 million m^3	International reporting and practice
Specified dam (Dam Safety Act, 2021)	Height above 15 m; or 10–15 m meeting at least one design condition in the Act (such as reservoir capacity, crest length, spillway discharge, or difficult foundation / unusual design)	Legal obligations: inspections, instrumentation, EAP, comprehensive safety evaluation, records
Hazard potential (high / significant / low)	Based on consequences downstream if the dam fails — mainly population and property at risk	Sets alert strictness, inspection frequency and EAP scope

Confirm the exact wording of the specified-dam criteria and the hazard classification method against the Act and the current NDSA / CWC guidelines.

3. How dams fail — failure modes

Monitoring exists to catch these mechanisms early. For embankment dams, overtopping and internal erosion are the two leading historical causes of failure; for concrete dams, foundation problems and overtopping dominate. A structured Potential Failure Mode Analysis (PFMA) for each dam decides what to monitor.

Failure mode	What happens (plain English)	Dam types	Early warning signs	Parameters that detect it
Overtopping	Water flows over the crest; embankments wash away quickly	All; embankments most vulnerable	Reservoir rising faster than planned, gates not opening, debris blocking spillway	Reservoir level and rate of rise, inflow forecast, rainfall, gate position, power for hoists
Internal erosion / piping	Seeping water carries soil particles out, forming a growing pipe	Embankment; also foundations of concrete dams	Rising seepage at constant level, cloudy (turbid) seepage, new wet spots, boils, sinkholes	Seepage flow, turbidity, piezometers, DTS fibre temperature, visual inspection
Slope instability	Part of the embankment slides	Embankment	Bulging, cracks along slope, high pore pressure, inclinometer movement	Piezometers, inclinometers, surface movement, rapid drawdown rate
Sliding (concrete)	Dam block moves along its foundation	Gravity, buttress	Uplift above design, drains clogged, downstream movement	Uplift piezometers, drain flow, pendulum, base joint meters
Overturning / overstress	Excessive tension or compression	Gravity, arch	Cracking at heel or toe, abnormal strain	Strain / stress meters, crack meters, pendulums
Foundation / abutment failure	Rock wedge or weak layer moves	Arch especially; all dams	Abutment movement, new seepage paths, rising foundation pressures	Extensometers, piezometers, seepage per zone, GNSS
Cracking and joint opening	Cracks let water in, raise pressure, start erosion	All	Growing crack width, new leakage	Crack / joint meters, seepage, visual
Concrete deterioration (AAR, leaching)	Concrete swells (alkali–aggregate reaction) or dissolves	Concrete, masonry	Irreversible upstream–downstream drift, rising crest, white deposits	Pendulums, levelling, seepage chemistry (calcium, pH)
Liquefaction (earthquake)	Saturated loose soil loses strength in shaking	Embankment, tailings	Sand boils, slumping after an earthquake	Accelerographs, piezometers, post-earthquake inspection
Spillway / gate failure	Spillway erodes or gates cannot open	All gated dams	Cavitation damage, failed gate tests, slab uplift	Gate position, hoist load, vibration, chute underdrain pressure
Rapid drawdown failure	Fast reservoir lowering leaves trapped water in the upstream slope	Embankment	Drawdown faster than the design rate	Reservoir level rate of fall, upstream piezometers
Reservoir rim landslide	Hillside slides into the reservoir, creating a wave	All	Rim cracks, movement after rain or drawdown	GNSS / InSAR on rim, rainfall, inclinometers

4. Health monitoring parameters

Parameters fall into nine groups. The digital tool uses these groups for the Dam Health Index and for linking each sensor to the failure modes it watches.

Group	Parameter	Unit	Why it is measured
A. Loads (hydraulic)	Reservoir level, tailwater level, rate of rise / fall	m, m/day	The main load on the dam; every response is read against it
	Inflow, outflow, spillway discharge	m^3/s	Flood handling, routing, overtopping risk
	Rainfall	mm	Explains seepage and pore pressure spikes; flood forecasting
	Wave height, wind, ice (where relevant)	m, km/h	Freeboard and crest protection
B. Seepage	Seepage flow (total and by zone)	L/min, L/s	Most important single indicator of water passing through or under the dam
	Turbidity, sediment in seepage	NTU, visual	Soil being carried out = internal erosion
	Seepage chemistry (pH, conductivity, calcium, dissolved solids)	—, µS/cm, mg/L	Leaching of concrete / grout or dissolving foundation rock
	Drain and relief-well flows	L/min	Drain effectiveness

Group	Parameter	Unit	Why it is measured
C. Water pressure	Pore water pressure (embankment, foundation)	kPa or m of water	Effective stress, slope stability, seepage path
	Uplift pressure under concrete dams	kPa or m	Reduces sliding resistance; compared with design uplift diagram
	Phreatic (saturation) line	m above MSL	Position of the seepage surface in embankments
D. Deformation — surface	Crest settlement and horizontal movement	mm	Overall behaviour, settlement, slope movement
	Pendulum (plumb / inverted) displacement	mm	Concrete dam deflection and foundation movement
	Tilt	arc-second / µrad	Block rotation
	Reservoir rim movement (GNSS, InSAR)	mm, mm/year	Landslide watch
E. Deformation — internal	Inclinometer profile	mm	Shear zones / slip surfaces inside embankments
	Settlement cells, extensometers	mm	Internal settlement, foundation compression, abutment movement
	Joint and crack opening (incl. CFRD perimeter joint 3-D)	mm	Opening, sliding, settlement across joints
F. Stress, strain, temperature	Concrete strain, no-stress strain, stress	µε, MPa	Overstress, thermal and AAR effects
	Concrete and water temperature	°C	Explains seasonal movement; mass-concrete cracking
	Total earth pressure	kPa	Stress in embankment core (arching, hydraulic fracture)
G. Seismic	Peak ground acceleration, response spectra	g	Earthquake loading; triggers special inspection
H. Operational / mechanical	Gate position, hoist load, gate vibration, standby power status	%, kN, mm/s, on/off	Ability to pass floods safely
I. Visual and remote	Cracks, wet areas, boils, vegetation, erosion (inspection, CCTV, drones)	observations, images	Things instruments cannot see; confirmation of instrument signals
	Reservoir sedimentation (bathymetry)	Mm^3 lost	Storage loss, intake blockage, silt load on dam

5. Codes, standards and guidelines

These are the documents an Indian dam monitoring programme normally relies on. Titles are given as commonly cited. The editions in force and the specific clauses relied on must be confirmed from the official publications; the digital tool records this confirmation against each standard and blocks approvals until it is done.

5.1 Law and regulation

Document	What it covers
Dam Safety Act, 2021 (Government of India)	Specified dams; National Committee on Dam Safety, National Dam Safety Authority (NDSA), State Committees and State Dam Safety Organisations (SDSO); owner duties: pre- and post-monsoon inspections, instrumentation and hydro-meteorological stations, records, Emergency Action Plan, periodic comprehensive dam safety evaluation by an independent panel
NDSA regulations and notifications	Detailed procedures issued under the Act (formats, timelines, reporting)

5.2 Design standards (BIS)

Standard	Subject
IS 6512	Criteria for design of solid gravity dams
IS 1893 (Part 1)	Earthquake-resistant design — seismic zones; site-specific parameters for river valley projects are separately approved
IS 7894	Stability analysis of earth dams
IS 8826	Design of large earth and rockfill dams

Standard	Subject
IS 12169	Design of small embankment dams
IS 10635	Freeboard requirements in embankment dams
IS 9429	Drainage systems for gravity dams, foundations and abutments
IS 456	Plain and reinforced concrete

5.3 Hydraulic structures (BIS)

Standard	Subject
IS 11223	Fixing spillway capacity (inflow design flood)
IS 6934	Hydraulic design of high ogee overflow spillways
IS 4997	Hydraulic-jump stilling basins
IS 7365	Bucket-type energy dissipators
IS 4623 / IS 4622	Radial gates / fixed-wheel gates
IS 6938 / IS 10210	Rope-drum hoists / hydraulic hoists
IS 11639	Penstocks
IS 7396	Surge tanks
IS 4880	Tunnels conveying water
IS 9761 / IS 11388	Hydropower intakes / trash racks

5.4 Instrumentation and monitoring (BIS)

Standard	Subject
IS 7436 (Part 1)	Types of measurements and choice of instruments — earth and rockfill dams
IS 7436 (Part 2)	Types of measurements and choice of instruments — concrete and masonry dams
IS 8282	Installation, observation and maintenance of pore-pressure instruments in earth dams
IS 4967	Seismic instrumentation for river valley projects
IS 10334	Cables for measuring devices in concrete and masonry dams

IS 8282, IS 4967 and IS 10334 titles and parts in particular should be checked against the current BIS catalogue.

5.5 Central Water Commission (CWC / DRIP) guidelines

Guideline	Use
Guidelines for instrumentation of large dams	Instrument selection, layout, reading frequency, data handling
Guidelines for safety inspection of dams	Pre- / post-monsoon and special inspections, checklists
Guidelines for developing Emergency Action Plans	Emergency levels, notification, inundation maps
Guidelines for assessing and managing risks associated with dams	Risk assessment, PFMA
Guidelines on design flood / inflow design flood	Flood standards for dam safety
Manual for rehabilitation of large dams; O&M manual guidelines	Remedial works, routine operation

Exact titles and years of the CWC / DRIP series to be listed from the CWC website at the time of use.

5.6 International references

Document	Use
ICOLD bulletins on dam surveillance and monitoring	Principles, behaviour models, data management
USACE engineering manuals on instrumentation of embankment and concrete structures	Instrument practice, thresholds
USBR Design of Small Dams and related manuals	Design and safety practice
FEMA Federal Guidelines for Dam Safety; EAP guidelines	Emergency planning, hazard classification

6. IoT sensors and instruments

Most modern dam instruments are "smart-ready": they produce an electrical signal that a data logger can read automatically. The table gives what each sensor measures, how, and what raw signal reaches the logger. Ranges are typical market values for selection, not specifications.

Instrument / IoT sensor	Measures	Working principle	Raw output to logger	Engineering unit	Typical range / resolution	Used in
Vibrating-wire (VW) piezometer	Pore / uplift water pressure	Water pressure deflects a diaphragm, changing a tensioned wire's frequency	Frequency (Hz) or "digits" (f^2/1000) + thermistor resistance (Ω)	kPa, m of water	0.35–3.5 MPa; ~0.025% FS resolution	All dams
Standpipe / Casagrande piezometer + level sensor	Water level in a pipe	Open pipe; manual dip meter or submersible pressure transducer	Manual reading, or 4–20 mA / Modbus	m above MSL	Depth of pipe; ±1 cm	Embankments, foundations
Uplift pressure gauge (gallery)	Uplift under concrete dam	Pipe from foundation to Bourdon gauge or transducer in the gallery	Manual or 4–20 mA	kPa, m	To full reservoir head	Gravity, buttress
V-notch / rectangular weir + level sensor	Seepage flow	Water level above the notch converted by the weir equation	Ultrasonic / radar / pressure level: 4–20 mA, Modbus, SDI-12	L/min, L/s	0–several hundred L/s; ±1 mm head	All dams
Electromagnetic / ultrasonic flow meter	Drain or pipe flow	Faraday / transit-time principle	Pulse, 4–20 mA, Modbus	L/min	Pipe-size dependent	Drains, relief wells
Turbidity sensor	Cloudiness of seepage	Light scattering (nephelometric)	4–20 mA, Modbus RS-485	NTU	0–1 000 (to 4 000) NTU	Seepage outfalls
Multi-parameter water quality probe	pH, conductivity, temperature	Electrochemical sensors	Modbus / SDI-12	pH, µS/cm, °C	Standard ranges	Seepage of concrete / masonry dams
Plumb / inverted pendulum + optical reader	Horizontal deflection (2 axes)	Wire hangs vertically; optical or capacitive reader tracks position	Digital (RS-485), 4–20 mA	mm	±25–50 mm; 0.01–0.1 mm	Gravity, arch
In-place inclinometer (MEMS / servo)	Tilt along a borehole → lateral displacement profile	Accelerometers sense gravity angle in each segment	Digital (RS-485 / SDI-12) tilt per sensor	mm (cumulative), mm/day	±10–30°; ~0.01 mm/m	Embankments, slopes, rim
Probe inclinometer (manual)	Same, by traversing probe	Reading at fixed depth intervals	Manual readout files	mm	Same	Embankments
Rod / borehole extensometer	Axial movement between anchors	VW or potentiometric displacement transducer	Hz / mV	mm	25–150 mm; 0.01–0.1 mm	Foundations, abutments
Settlement cell / hydraulic overflow cell	Internal settlement	Liquid-level or pressure difference to a reference	Hz or mA	mm	Up to several m; mm resolution	Embankment, CFRD
Joint meter / crack meter (incl. 3-D)	Opening, sliding, settlement at joints / cracks	VW or potentiometric transducer	Hz / mV	mm	12–100 mm; 0.01–0.025 mm	Concrete joints, CFRD perimeter joint
Tiltmeter / clinometer	Rotation of a surface	MEMS or electrolytic level	Digital / mA	arc-sec, µrad	±1–15°; arc-second resolution	Concrete blocks, structures
VW strain gauge / no-stress strain meter	Concrete strain	Wire frequency changes with strain	Hz + thermistor	µε	±1 500–3 000 µε; ~1 µε	Concrete dams
Stress meter / total pressure cell	Stress in concrete / earth pressure	Fluid-filled pad + VW transducer	Hz	MPa, kPa	Up to several MPa	Concrete, embankment core
Thermometer / thermistor string	Concrete, water, air temperature	Resistance changes with temperature	Ω	°C	−20 to +80 °C; 0.1 °C	Concrete dams, reservoir
Distributed fibre-optic sensing (DTS / DSS)	Temperature / strain along a cable (km)	Light scattering in optical fibre	Interrogator data file	°C, µε along length	Metre spatial resolution	Embankment seepage detection, CFRD
GNSS station	3-D position of points	Satellite positioning (static / RTK)	NMEA / RINEX via interrogator	mm	~2–5 mm horizontal, ~5–10 mm vertical (static)	Crest, abutments, rim
Robotic total station + prisms	Coordinates of many points	Automated angle and distance	Digital	mm	~1 mm + 1 ppm	Crest, downstream face

Instrument / IoT sensor	Measures	Working principle	Raw output to logger	Engineering unit	Typical range / resolution	Used in
		measurement				
Satellite InSAR (service)	Ground movement over wide areas	Radar phase change between passes	Processed data set	mm/year	mm-level trends	Dam body, rim, abutments
Reservoir / tailwater level sensor	Water level	Radar, pressure, or float encoder	4–20 mA, Modbus, SDI-12	m above MSL	±5–10 mm	All
Tipping-bucket rain gauge, weather station	Rain, temperature, humidity, wind	Mechanical tips; electronic sensors	Pulses; digital	mm, °C, %, km/h	0.2–0.5 mm per tip	All
Strong-motion accelerograph	Earthquake acceleration	Force-balance / MEMS accelerometer, event-triggered	High-rate digital records (100–200 samples/s)	g, cm/s²	±2 g	Crest, base, free-field
Gate position encoder, hoist load cell, vibration sensor	Gate opening, lifting force, vibration	Absolute encoder; strain-gauge load cell; accelerometer	Digital / mA	%, m, kN, mm/s	Equipment-specific	Gated spillways
CCTV / AI camera, drone survey	Visual condition, wet areas, cracks	Imaging + image analytics	Video / images	Observations	—	All

6.1 Where to install — typical layouts

Dam type	Typical layout (minimum)
Concrete / masonry gravity	Monitoring blocks at maximum section and on each abutment; foundation-gallery uplift piezometers across the base (heel, drain line, mid-base, toe); seepage weirs per gallery zone; drain-flow measurement; pendulum (plumb + inverted) in key blocks; joint meters across contraction joints; thermometers and strain gauges in a few blocks; crest survey points; reservoir level; accelerographs at crest and base
Arch	As gravity plus abutment extensometers and piezometers, multiple pendulums at crown and quarter points, dense temperature measurement, crown and abutment strain
Embankment (earth / ECRD)	Three to five cross sections (maximum section, abutments, any foundation anomaly): piezometers in foundation, core, filters and downstream shell; inclinometers in downstream slope and near abutments; settlement cells; total pressure cells in high cores; seepage weirs separating left, right and central zones; turbidity at each weir; crest monuments every ~50–100 m; downstream toe observation
CFRD	Perimeter-joint 3-D meters, face-slab joint meters and strain, rockfill settlement (hydraulic overflow cells), crest and face survey, leakage weir downstream (often the single best health indicator)
Barrage / weir	Uplift pressure under the floor at several points, downstream scour survey, gate instrumentation, piping watch at abutments

> Redundancy: critical locations should have at least two independent measurements (for example a VW piezometer and a standpipe), because instruments fail over the decades and a single silent sensor must not hide a problem.

7. What data is received and how

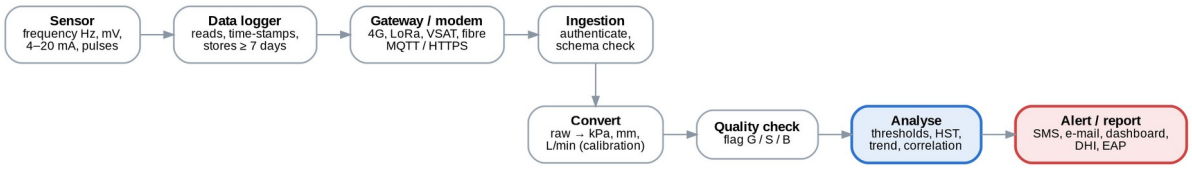


Figure 7.1 — Data chain from sensor to alert

Stage	Data form	Example	Checks
Sensor output	Raw electrical signal	VW piezometer: 2 873.4 Hz, thermistor 3 012 Ω	Signal present, within sensor range
Logger record	Raw value + timestamp + channel	"2026-10-04 18:00, ch 7, 8 255.6 digits, 27.1 °C"	Clock synchronised (NTP / GNSS), battery
Transmission	Message (JSON over MQTT / HTTPS) or file (CSV)	Message with message ID, gateway ID, readings array	Authentication, duplicates, completeness
Engineering value	Converted using calibration	Pressure = $G\cdot(R_0 - R_1) + K\cdot(T_1 - T_0)$ → 198.4	Calibration version valid at reading

Stage	Data form	Example	Checks
	sheet	kPa	time
Validated value	Value + quality flag	198.4 kPa, flag G	Range, spike, flatline, gaps
Derived values	Indicators	Piezometric level = tip elevation + p / 9.81; seepage L/min from weir head	Formula documented and tested
Analysed result	Status, anomaly, trend, forecast	Level 1 Alert; 20 days to alarm	Engineer review

7.1 Minimum fields every reading must carry

Field	Purpose
Dam ID, sensor ID, parameter code	Which instrument and what quantity
Observed date-time with time zone	When it was measured (not when it arrived)
Value and unit (and raw value where available)	The measurement; raw value allows re-conversion after a calibration correction
Logger status, battery, signal strength	Instrument and communication health
Message / sequence ID	Detect duplicates and missing messages
Reading method (automatic / manual)	Manual readings carry observer name and photo

7.2 How often to read

Frequency depends on the phase of the dam's life. The first filling of the reservoir is the most dangerous period — many historical failures occurred then — so readings are most frequent. The values below are typical starting points; the dam's instrumentation report and CWC guidance set the final schedule. Automated sensors can read far more often at no extra cost; the platform then summarises to the required interval.

Instrument	Construction	First filling	First ~5 years	Long-term operation	After an event
Reservoir level, rainfall	Daily	Hourly	Hourly–daily	Hourly–daily (automatic)	Continuous
Piezometers / uplift	Weekly	Daily	Weekly	Weekly–monthly (manual); 1–6 h (automatic)	Immediately, then daily
Seepage and turbidity	—	Daily	Weekly	Weekly (manual); hourly (automatic)	Immediately, then several times a day
Pendulums, joint / crack meters	Weekly	Daily	Weekly	Monthly (manual); 6 h (automatic)	Immediately
Inclinometers, settlement	Weekly	Weekly	Monthly	Quarterly–half-yearly (manual); daily (automatic)	Immediately
Survey (GNSS / total station)	Monthly	Weekly–monthly	Quarterly	Half-yearly–yearly; continuous with GNSS	Immediately
Accelerographs	—	Event-triggered	Event-triggered	Event-triggered	Download at once
Visual inspection	Continuous	Daily	Weekly	Routine + pre- / post-monsoon	Special inspection

7.3 Communication options

Option	Best for	Note
4G / LTE modem	Sites with mobile coverage	Low cost; plan for outages
LoRaWAN	Many low-power sensors spread around the dam, to a local gateway	Long battery life; low data rate
Fibre / Ethernet in galleries	Dense instrumentation in concrete dams	Most reliable inside the dam
VSAT / satellite	Remote sites without mobile coverage	Higher cost; good backup
Radio (UHF / VHF)	Linking remote stations to the dam control room	Licensing required

> Every logger should store at least seven days of data locally, and every critical channel should have a backup path (second network or manual reading), so that a communication outage during a flood never means blind monitoring.

8. How dam safety monitoring is done with the data

Raw numbers become safety decisions through layers of checks. Each layer catches something the others miss.

Layer	What it does (plain English)	Example
1. Validation	Reject impossible values, flag stuck or noisy sensors	A seepage reading of 950 L/min from a weir with 400 L/min capacity is flagged bad
2. Absolute thresholds	Compare with fixed limits set from design and experience	Uplift above the design uplift → alarm
3. Expected behaviour (model)	Compare with what this dam normally does at this reservoir level and season	Piezometer 4 m higher than its usual value for the same reservoir level
4. Trend and rate of change	Is it moving steadily or accelerating?	Crest settlement speeding up instead of slowing down
5. Correlation	Does it follow its cause (reservoir, rain, temperature)?	Seepage rising while reservoir is steady = suspicious
6. Corroboration	Do related instruments agree?	Seepage up + turbidity up + nearby piezometer up = strong signal
7. Visual check	Does the site confirm it?	Wet patch found at the location the instruments point to
8. Engineering judgement	Qualified engineer decides cause, risk and action	Lower reservoir, clean drains, special investigation

8.1 The behaviour model (HST) explained

Most dam readings move for three ordinary reasons: the water level (hydrostatic, H), the time of year (seasonal temperature, S), and slow ageing (time, T). An HST model learns each instrument's normal pattern from a few years of good data, like learning a person's normal blood pressure at rest and after exercise. The model then predicts today's expected reading; the difference between measured and predicted (the residual) is what matters.

$$\text{Reading} = a_0 + a_1 \cdot h + a_2 \cdot h^2 + a_3 \cdot h^3 + a_4 \cdot h^4 \quad \text{(water level effect, } h = \text{normalised level)}$$
$$+ b_1 \cdot \sin s + b_2 \cdot \cos s + b_3 \cdot \sin 2s + b_4 \cdot \cos 2s \quad \text{(season, } s = 2\pi \times \text{day of year} / 365)$$
$$+ c_1 \cdot t + c_2 \cdot \ln(1 + t) \quad \text{(time / ageing trend)}$$
$$\text{Residual} = \text{measured} - \text{predicted}\ ; \quad \text{alert when } |\text{residual}| > k \cdot \sigma \text{ of past residuals } (k \approx 2\text{–}3)$$

- A rising time term (c_1) shows irreversible behaviour — creep, AAR swelling, drain clogging or progressive erosion — that would be hidden inside normal seasonal swings.
- Requires roughly two to three years of good data to fit reliably; until then, use statistical bands and design limits.
- Thermal variant (HTT) uses measured concrete temperatures instead of the seasonal sine terms where available.

8.2 Other analysis methods

Method	Use
Statistical band (rolling mean ± k·σ)	Simple anomaly detection before a model exists (used in DAM MASTER v1.0)
Historical envelope at the same reservoir level	"Has this instrument ever read this high at this level?"
Rate-of-change limits	Catch fast changes even when below absolute limits
Back-analysis with numerical models (FE)	Compare measured behaviour with predicted design behaviour; update material properties
Machine learning (e.g. gradient boosting, LSTM)	Better prediction of complex behaviour; must remain explainable and validated against HST
Multi-sensor rules / failure-mode indicators	Combine signals linked to one failure mode (e.g. piping index)

9. What changes in the data can mean

The tables show typical interpretations. The words "normal", "warning" and "critical" describe the pattern of change, which is often more important than the size. Any numbers are illustrative and must be replaced by values from the dam's design and history.

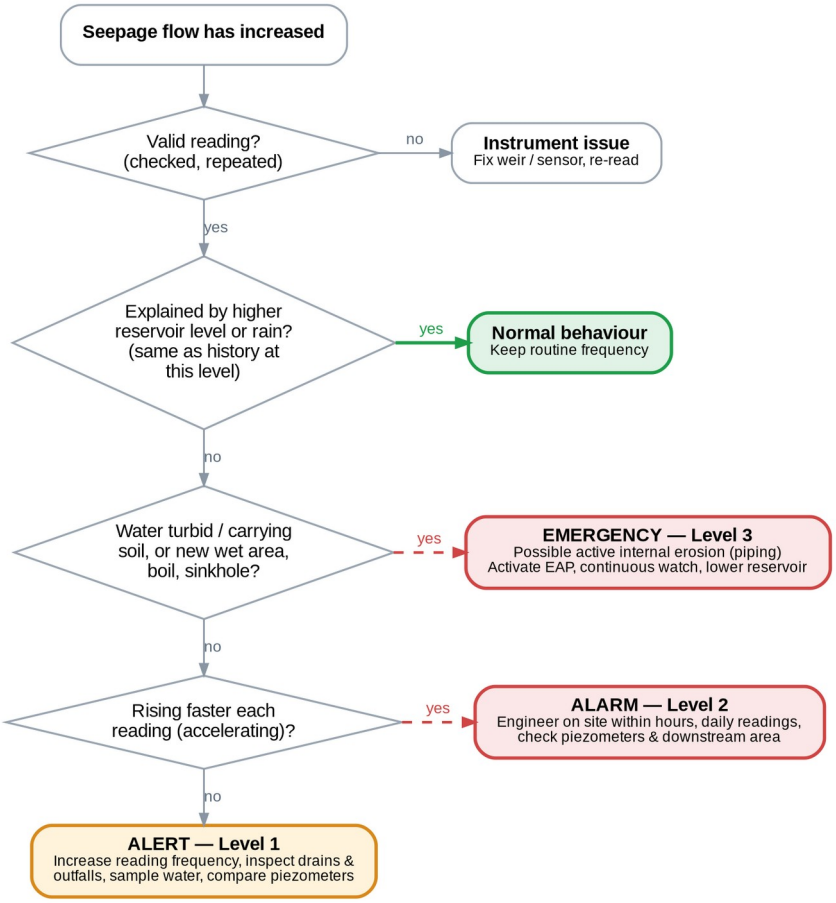


Figure 9.1 — Interpreting an increase in seepage

9.1 Seepage, turbidity and drains

Observation	Normal pattern	Warning sign	Critical sign	Possible cause	Action
Seepage flow	Rises and falls with reservoir level along a repeatable curve	Higher than usual for the same level; steady rise at constant level	Rapid or accelerating rise; new concentrated leak	Cracking, joint opening, grout curtain deterioration, internal erosion	Increase readings, inspect, compare piezometers, water sample
Turbidity	Clear water (low, stable NTU)	Occasional cloudiness after rain or maintenance	Persistent or increasing cloudiness, visible soil in weir box	Soil being transported — internal erosion	Treat as possible piping: immediate inspection, EAP readiness
Sudden drop in seepage	—	Flow drops without reservoir fall	Drop plus rising pore pressure behind	Drains clogging, flow finding another path	Check drains; pressures may be building up
Seepage chemistry	Stable pH, conductivity	Rising calcium / conductivity	White deposits growing, rising dissolved solids	Leaching of concrete / grout; dissolving foundation rock	Chemical analysis, check strength / permeability
Downstream wet areas, boils, sinkholes	None	New damp area	Boil carrying sand, sinkhole, cloudy spring	Piping, concentrated leak	Emergency inspection; EAP if sediment is moving

9.2 Water pressure (piezometers and uplift)

Observation	Normal pattern	Warning sign	Critical sign	Possible cause	Action
Embankment pore pressure	Follows reservoir with a lag; stable relation	Higher than history for the same level; faster	Downstream piezometers rising	Core / filter defect, blocked drain, new	Slope stability check, increase readings, check drains

Observation	Normal pattern	Warning sign	Critical sign	Possible cause	Action
		response	toward the slope surface	seepage path	
Uplift under concrete dam	Matches design uplift diagram (reduced at drain line)	Above design at drain line; drain flow falling	Uplift close to full reservoir head	Drains clogged, curtain deteriorated, foundation cracking	Clean / re-drill drains; recompute sliding factor of safety
Very fast response to reservoir	—	Lag shortening over time	Pressure jumps with level almost instantly	Direct hydraulic connection (crack or pipe) to the reservoir	Investigate; possible grouting or remedial work
Rapid drawdown	—	Lowering faster than design rate	Upstream pore pressures stay high while level falls fast	Trapped water in upstream slope	Slow the drawdown; slope check

9.3 Movement and deformation

Observation	Normal pattern	Warning sign	Critical sign	Possible cause	Action
Concrete dam deflection (pendulum)	Moves downstream with high water and summer heat, back again; repeatable loop	Loop shifting gradually downstream year by year	Sudden step or accelerating drift	Creep, AAR swelling, foundation movement, sliding	HST analysis, foundation investigation
Embankment crest settlement	Large after construction, then slows (roughly with log of time)	Settlement rate not decreasing	Accelerating settlement, local depression, crest cracks	Internal erosion, collapse settlement on wetting, slope movement	Survey more often, inspect, check freeboard
Inclinometer profile	Small uniform movement	Movement concentrated at one depth	Growing shear zone, mm/day and accelerating	Developing slip surface	Emergency evaluation; lower reservoir if needed
Joint / crack opening	Seasonal opening and closing	Progressive opening year on year	Rapid opening or new cracking with leakage	Differential movement, overstress	Structural assessment, sealing
CFRD perimeter joint	Small settlement and opening after first filling	Continued growth	Opening beyond water-stop capacity with leakage rise	Slab movement, rockfill settlement	Leakage control, joint repair
Reservoir rim	No movement	Slow movement after rain or drawdown	Accelerating movement	Landslide developing	Restrict drawdown, warning to downstream / reservoir users

9.4 Loads, earthquakes and equipment

Observation	Warning sign	Critical sign	Action
Reservoir level	Above rule curve; rate of rise above forecast	Approaching MWL with gates already fully open; forecast above top of dam	Pre-release per rules; flood watch; EAP if overtopping possible
Earthquake	Felt shaking or recorded acceleration above the trigger set in the EAP	Strong shaking; new cracks, seepage or settlement after the event	Special post-earthquake inspection and readings of all instruments
Spillway gates	Failed operation test; hoist overload; vibration	Gate cannot open during a flood; no standby power	Repair; alternative release; EAP
Instrument / data health	Sensor offline, battery low, data gap	Critical sensors silent during a flood or high level	Restore; manual readings until fixed

10. Thresholds and alerts — when alerts must go

10.1 How thresholds are set

Method	How	Strength	Weakness
Design-based	From design calculations: design uplift, allowable movement, pore pressure assumed in stability analysis	Linked directly to safety margin	Design assumptions may differ from real behaviour
Statistical	Mean ± k·σ of a stable baseline, or HST residual limits	Adapts to the real dam	Needs good history; may normalise slow deterioration
Historical envelope	Highest / lowest ever recorded at the same reservoir level	Easy to explain	Only as good as the record
Rate of change	Maximum allowed change per day / week	Catches fast developments early	Sensitive to noise — needs validation

Method	How	Strength	Weakness
Failure-mode based (PFMA)	Value at which a specific failure mode becomes credible	Ties alerts to consequences	Needs expert workshop per dam

Good practice combines them: Level 1 from statistics or history (unusual), Level 2 from design limits or accelerating trends (unsafe trend), Level 3 from failure-mode thresholds or visual evidence of a developing failure (danger).

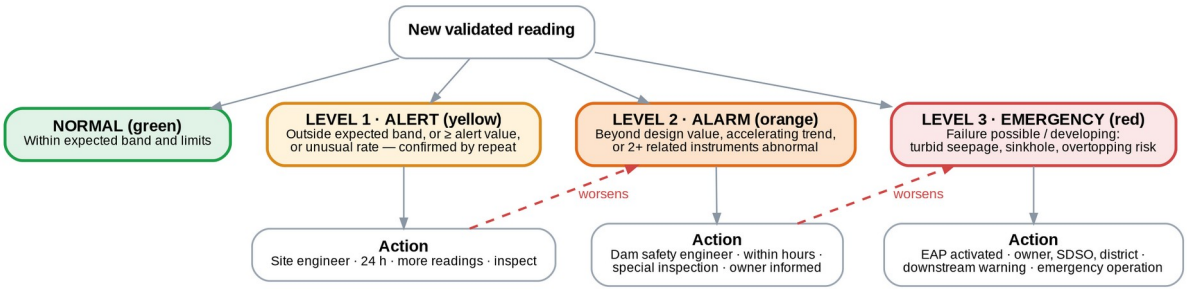


Figure 10.1 — Alert levels and response

10.2 The four-level scheme

Level	Meaning	Typical triggers	Who is told, how fast (indicative)
Normal (green)	Behaving as expected	All readings within expected bands and below limits	Routine reports
Level 1 — Alert (yellow)	Unusual, needs attention	Outside statistical band, above alert value, unusual rate, confirmed by repeat reading	Site / instrumentation engineer; in-app + e-mail; acknowledge within 24 h
Level 2 — Alarm (orange)	Unsafe trend or design limit reached	Above design value, accelerating trend, two or more related instruments abnormal, visual confirmation	Dam safety engineer and owner; SMS + e-mail; acknowledge within ~2 h; special inspection
Level 3 — Emergency (red)	Failure possible or developing	Turbid seepage increasing, sinkhole, slope sliding, overtopping imminent, major earthquake damage	EAP: owner, SDSO, district administration, downstream warning; all channels; minutes

Response times and contacts are set in each dam's Emergency Action Plan; values here are placeholders to be confirmed.

10.3 Illustrative alert rules per parameter

Example rule set for a new dam programme. Every number is a placeholder showing the form of the rule; the dam safety engineer replaces it with dam-specific values.

Parameter	Level 1 — Alert (example)	Level 2 — Alarm (example)	Level 3 — Emergency (example)
Seepage flow	> expected for current level by 2σ, or +20% over 7 days at steady level	> design / historical maximum, or accelerating 3 readings in a row	Sudden large rise with turbidity, or new concentrated leak
Turbidity	Above baseline (e.g. > 2× normal) after rain	Persistent above baseline at steady conditions	Visible soil / rising NTU with rising seepage
Piezometer (embankment)	Above historical envelope for current level	Above value assumed in slope stability design	Downstream piezometers near ground surface, or rising fast with movement
Uplift (concrete)	Above expected HST value by 2σ	Above design uplift at drain line	Approaching full reservoir head with movement
Pendulum / crest movement	Outside HST band	Irreversible drift above set mm/year, or step change	Accelerating movement with cracking or leakage
Inclinometer	Movement localised at one depth	Rate above set mm/day	Accelerating rate
Reservoir level	Above rule curve / rising faster than plan	Within set margin of MWL	Forecast above top of dam
Earthquake	Recorded PGA above inspection trigger	Strong shaking near the dam	Damage observed
System health	Sensor offline > 3 intervals; battery low	Critical sensor offline during high level or flood	—

10.4 Rules that keep alerts trustworthy

- **Validate first:** never alert on a value flagged bad; confirm a Level 2 or 3 trigger with a repeat reading or a second instrument (unless the evidence is visual and obvious).
- **Persistence:** require the condition for N consecutive readings for Level 1, to avoid noise.
- **Rate and absolute together:** a fast change below the limit can matter more than a slow one above it.
- **Corroboration raises the level:** two or more related instruments abnormal for the same failure mode escalates one level.
- **No silence:** a sensor that stops reporting is itself an alert; critical sensors offline during floods escalate.
- **Acknowledge and escalate:** unacknowledged alerts escalate automatically up the contact list.
- **Close with a reason:** confirmed, false alarm or instrument fault — this feedback tunes thresholds.
- **Planned work:** maintenance windows suppress alerts only for named sensors and times, and are logged.

11. Inspections, EAP and records

Activity	When	What it covers
Routine inspection	As per O&M manual (daily–weekly by operators)	Visual walk-over, seepage points, gates, instruments
Pre-monsoon inspection	Every year before monsoon (Dam Safety Act)	Spillway and gates ready, drains, instruments, communication, EAP contacts
Post-monsoon inspection	Every year after monsoon (Dam Safety Act)	Damage from floods, erosion, new seepage, instrument behaviour during high levels
Special inspection	After earthquake, major flood, rapid drawdown, Level 2 / 3 alert	Targeted at the triggering event
Comprehensive dam safety evaluation	Periodically by an independent panel (Dam Safety Act)	Full review: hydrology, structure, instrumentation data, risk
EAP drills	Periodically	Test notification chain and downstream warning

Instrument data and inspection observations must be read together: an instrument shows where to look; the inspection confirms what is happening. The digital tool links every inspection finding to the nearest instruments and every alert to an inspection task.

12. Further capabilities worth adding

Capability	What it adds
Potential Failure Mode Analysis (PFMA) register	Lists each credible failure mode per dam with its indicators and thresholds — the backbone of monitoring design
Risk-informed safety	Combines probability of failure modes with downstream consequences; prioritises spending across a portfolio
Satellite InSAR	Millimetre movement of dam, abutments and reservoir rim without field instruments; good for portfolios of older dams
Drones and AI image analysis	Crack, seepage-patch and vegetation detection on faces and slopes; repeatable 3-D surveys
Distributed fibre optics	Continuous seepage detection along the full length of embankments
Hydro-meteorological network and flood forecasting	Earlier warning of high inflows; pre-release decisions
Sedimentation monitoring (bathymetry)	Tracks storage loss and intake / outlet blockage
Digital twin	3-D model of the dam with live sensor data and design calculations in one place
Dam Health Index	One score per dam and per component for management and portfolio ranking
Downstream early warning	Sirens and SMS linked to EAP Level 3
Instrument life-cycle management	Calibration, maintenance, replacement planning; reliability statistics
Cyber security	Protects the monitoring and gate-control systems from tampering

13. Turning this into the digital tool

Each chapter becomes master data or a rule set in the DAM MASTER AI platform. Building these tables first makes the software configurable for any dam without code changes.

Knowledge (chapter)	Becomes in the tool	Status in DAM MASTER v1.0
Dam types (2)	Dam master: type, function, height, hazard, specified-dam flag	Partly (type, hazard, height)
Failure modes (3)	PFMA register linked to dams, instruments and alert rules	Planned
Parameters (4)	Parameter catalogue with units and groups	21 parameter codes
Codes (5)	Reference register with edition / clause verification	19 references, verification workflow
Sensors (6)	Instrument catalogue: type, raw output, conversion, location	Sensor master; conversions documented
Data (7)	Ingestion API, Excel template, quality rules, reading schedules	Excel import + 8 QC rules
Analysis (8)	Statistical bands now; HST models when 2–3 years of data exist	z-score + trend; HST planned
Meaning of changes (9)	Interpretation library shown in alerts ("possible causes / actions")	Partly (triage, piping rule)
Alerts (10)	Four-level rule engine: absolute, statistical, rate, persistence, corroboration	3-level thresholds + triage
Inspections & EAP (11)	Inspection schedules, findings, EAP contacts and drills	Findings register
Further capabilities (12)	InSAR, drones, fibre, forecasting connectors	Planned

13.1 Master data the tool needs per dam

1. Dam identity, type, function, hazard, levels (FRL, MWL, MDDL, crest), specified-dam status.
2. PFMA list: failure mode, location, indicators, related instruments.
3. Instrument list: type, location, elevation, raw output, calibration constants, reading schedule by phase.
4. Thresholds per instrument: Level 1–3 values or statistical rules, rate limits, persistence, direction, method used to set them, approver and date.
5. Alert routing: contacts per level, channels, acknowledgement times, escalation order (from the EAP).
6. Reference register: standards and guidelines with verified edition and clauses.

13.2 Recommended next development steps

7. PFMA register and failure-mode linking of sensors and alerts.
8. Four-level alert engine with rate-of-change, persistence and corroboration rules.
9. Reading-schedule engine by life phase (construction, first filling, operation, post-event).
10. HST model fitting per instrument once enough data exists, with residual-based alerts.
11. Server version (ASP.NET Core + PostgreSQL / TimescaleDB) with live MQTT ingestion.

Appendix — Glossary

Term	Meaning
FRL / MWL / MDDL	Full reservoir level / maximum water level / minimum draw-down level
Phreatic line	Top of the saturated zone inside an embankment
Piping / internal erosion	Soil particles washed out by seepage, forming a pipe
Uplift	Upward water pressure under a concrete dam
AAR	Alkali–aggregate reaction — chemical swelling of concrete
HST model	Hydrostatic–Season–Time statistical model of normal behaviour

Term	Meaning
PFMA	Potential Failure Mode Analysis
EAP	Emergency Action Plan
SDSO / NDSA	State Dam Safety Organisation / National Dam Safety Authority
NTU	Nephelometric Turbidity Unit — cloudiness of water
PGA	Peak ground acceleration during an earthquake
VW	Vibrating wire — common robust sensor technology
DTS	Distributed temperature sensing along an optical fibre
InSAR	Satellite radar interferometry for ground movement